

Yichao v2 Brightfield-to-Fluorescence and Early Prediction Plan

OrganoidAgent

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1 Purpose

The first objective is to prepare a clean Yichao v2 instance-pair database and reproduce the successful v1 brightfield-to-fluorescence result on the new data. The second objective is to extend the model from same-day prediction to early prediction:

$$\mathbf{B}_d \mapsto \mathbf{F}_d \quad \text{and} \quad \mathbf{B}_k \mapsto \mathbf{F}_d, \quad k < d.$$

The early-prediction model should be encouraged to predict later fluorescence from earlier brightfield morphology, but it should not be over-penalized when the earliest morphology genuinely lacks enough information.

2 Current v2 Data State

The local v2 root is:

/home/lachlan/ProjectsLFS/OrganoidAgent/DATA-Yichao-v2/

Table 1: Yichao v2 LIF inventory after copying all five files. The first two files were previously extracted and previewed; the last three were copied and metadata-inspected only.

Folder	LIF	Day	Series	Fields	Z range	Channels	State
1_N39Rep_Globet_DF_D3N39Rep_Globet_DF_D33	if		40	10	20–49	2	extracted
2_N39Rep_Globet_DF_D4N39Rep_Globet_DF_D44	if		24	6	11–38	2	extracted
3_N39Rep_Globet_DF_D2N39Rep_Globet_DF_D22	if		15	15	1	7	copied
4_N39Rep_Globet_DF_D3N39Rep_Globet_DF_D33	1.lif		10	10	4–13	8	copied
5_N39Rep_Globet_DF_D3N39Rep_Globet_DF_D33	2.lif		12	12	12–31	8	copied

The important structural issue is that the v2 files are not homogeneous. The already-previewed D3/D4 files contain four named acquisition series per field, such as ALEXA488, DAPI, ALEXA647, and mCherry, with two internal channels per acquisition. The newly copied D2/D3_1/D3_2 files contain generic **Series###** names and seven or eight internal channels per series. Therefore, the new files require visual channel mapping before segmentation or training.

3 Channel Mapping

For the already-previewed named-acquisition D3/D4 files, the current practical mapping is:

input brightfield = ALEXA488 c1, target fluorescence = ALEXA488 c0.

The non-ALEXA488 channels should not be treated as independent target fluorescence unless there is a biological reason. They can be used for quality control or controls.

For D2/D3_1/D3_2, the correct mapping is unknown. A channel contact sheet must be made before any database is built. A safe rule is:

no confirmed channel map $\Rightarrow$ no segmentation database entry.

4 Dataset Preparation Pipeline

The preparation pipeline is intentionally separated from training.

1. Inspect metadata for every LIF file.
2. Extract all planes and group by field/series.
3. Create channel preview grids and manually confirm brightfield and fluorescence channels.
4. Segment the brightfield image, not the fluorescence image.
5. Crop paired brightfield and fluorescence instances with the same segmentation mask.
6. Store records in a v2 database with channel mapping, day, field, z policy, crop size, edge-padding flags, and split group.

The provided preparation command is:

```
bash analysis-tools/yichao_v2/run_yichao_v2_lif_prepare.sh
```

This command extracts and groups images, but it does not train and does not run segmentation.

5 Database Design

Each organoid crop should be represented as a single row:

$$r = (x_B, x_F, m, d, s, z, c_B, c_F, g),$$

where x_B is the brightfield crop, x_F is the fluorescence crop, m is the instance mask, d is day, s is source series/field, z is the z-plane or projection policy, c_B, c_F are channel identifiers, and g is the split group.

The split group is essential:

$$g = \text{donor/experiment} + \text{LIF} + \text{field/series} + \text{replicate}.$$

All rows with the same g must stay in the same train, validation, or test split.

6 Same-Day Model

The first trainable model should reproduce v1 behavior:

$$\hat{\mathbf{F}}_{i,d} = f_{\theta}(\mathbf{B}_{i,d}).$$

Use the robust v1 target style:

- background-suppressed continuous fluorescence target;
- optional positive mask target;
- balanced sampling for positive signal;
- 256 pixel crops as the first default.

A suitable same-day loss is:

$$\mathcal{L}_{\text{same}} = \lambda_F \left\| \hat{\mathbf{F}}_{i,d} - \mathbf{F}_{i,d}^{\text{clean}} \right\|_1 + \lambda_M \mathcal{L}_{\text{mask}}(\hat{\mathbf{M}}_{i,d}, \mathbf{M}_{i,d}^{\text{clean}}) + \lambda_E \left| \hat{E}_{i,d} - E_{i,d} \right|,$$

where $E_{i,d}$ is total or mean positive fluorescence.

7 Early-to-Later Prediction

The temporal extension is:

$$\hat{\mathbf{F}}_{i,d|k} = f_{\theta}(\mathbf{B}_{i,k}, \Delta), \quad \Delta = d - k, \quad k < d.$$

The horizon Δ is allowed because the requested prediction horizon is known. Future fluorescence and future brightfield are not allowed as inputs.

The full objective combines same-day and early-to-later tasks:

$$\mathcal{L} = \mathcal{L}_{\text{same}} + \sum_{k < d} w(d - k) \mathcal{L}_{\text{future}}(i, k, d).$$

The future loss can use the same target structure:

$$\mathcal{L}_{\text{future}} = \lambda_F \left\| \hat{\mathbf{F}}_{i,d|k} - \mathbf{F}_{i,d}^{\text{clean}} \right\|_1 + \lambda_M \mathcal{L}_{\text{mask}}(\hat{\mathbf{M}}_{i,d|k}, \mathbf{M}_{i,d}^{\text{clean}}) + \lambda_E \left| \hat{E}_{i,d|k} - E_{i,d} \right|.$$

8 Forgiven Early Horizons

Earlier brightfield may not contain enough information. A useful objective should reward early signal when it exists but avoid forcing impossible pixel-perfect prediction.

8.1 Fixed horizon weighting

A simple strategy is:

$$w(\Delta) = \exp(-\gamma\Delta).$$

This gives near-future prediction a larger weight and very early prediction a softer weight.

8.2 Learned uncertainty weighting

A better strategy is:

$$\mathcal{L}_\Delta = \frac{\mathcal{L}_{raw,\Delta}}{2\sigma_\Delta^2} + \log \sigma_\Delta.$$

The model learns which horizons are intrinsically uncertain. If D2 brightfield cannot reliably predict D4 fluorescence, the objective can still learn from D3-to-D4 while not destabilizing training.

9 No Future Leakage

No-leakage constraints are stricter than the model architecture.

1. Split by biological field/replicate before creating temporal pairs.
2. Keep all acquisitions and days from the same split group in the same split.
3. Use only data observed at or before day k as input for $k \rightarrow d$.
4. Do not normalize training inputs or tune thresholds using validation/test target distributions.
5. Fix channel mapping before held-out evaluation.
6. Select earliest-useful-day thresholds on validation only.

10 Earliest Useful Day

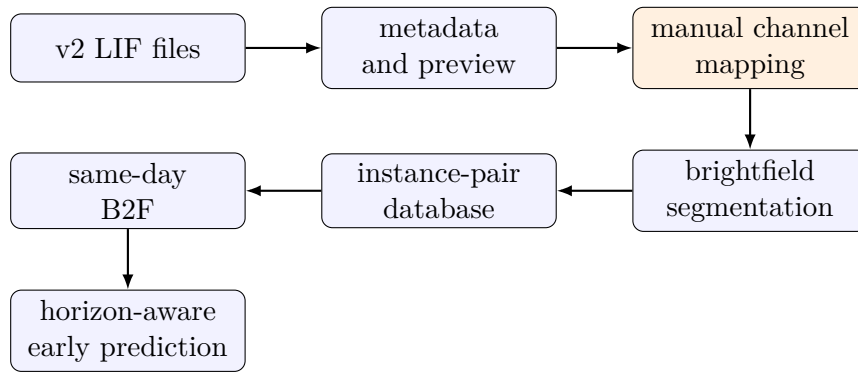
For each horizon $k \rightarrow d$, evaluate:

- positive-region Dice/F1;
- total fluorescence Pearson or Spearman correlation;
- organoid-level AUROC and AUPRC;
- calibration of total predicted expression.

The earliest useful day is:

$k^* = \min k$ such that validation metrics exceed the predefined threshold stably across replicates.

11 Recommended Flow



12 Current Decision

Do not train immediately. The safe next step is to extract and preview the newly copied D2/D3_1/D3_2 files, confirm their channel mapping, and only then build the v2 segmentation/instance database.